QUIZ 01 SOLUTIONS

Q1 Initial state ① : $p_o = 100\,kPa$, $V_1 = 0.3\,m^3$, $m = 2.5\,kg$

$$\therefore\ v_1 = \frac{V_1}{m} = 0.12\,m^3$$

@ 100 kPa, $v_f = 0.001043\,\frac{m^3}{kg}$, $v_g = 1.6941\,\frac{m^3}{kg}$.

Since, $v_f < v_1 < v_g \Rightarrow$ state ① is saturated mixture. (+1)

$$x_1 = \frac{v - v_f}{v_g - v_f} = 0.07026$$

$$\therefore\ u_1 = x_1 \cdot u_g + (1-x_1) u_f = 564.117\,\frac{kJ}{kg}$$ (+1)

State ②: When piston hits the wall, $V_2 = 0.625\,m^3$, $T_2 = 600^\circ C$

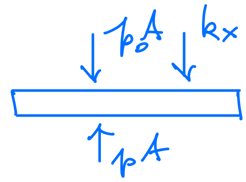
Since $T_2 > T_{crit.} \Rightarrow$ superheated state. (+1)

Also, $v_2 = \frac{V_2}{m} = \frac{0.625\,m^3}{2.5\,kg} = 0.25\,m^3/kg$ (+1)

From superheated table,

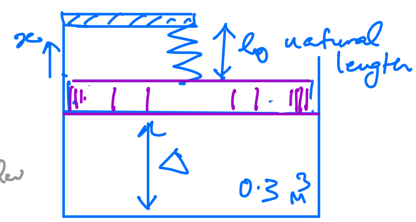
$T_2 = 600^\circ C$ and $v_2 = 0.25\,\frac{m^3}{kg}$ @ $p_2 = 1.6\,MPa$
$= 1600\,kPa$

To prove that a linear relation exist b/w p and V, we consider force balance of piston. Assuming piston is always in quasi-equilibrium. forces acting on it balance each other.



$$pA = p_o A + kx$$

or, $$p = p_o + \frac{kx}{A}$$ (+1)



It is given that @ $V = 0.3\,m^3$, spring is natural length.

piston Area; → volume of cylinder

$$(\Delta + x) \cdot A = V$$

$$\Rightarrow \qquad x = \left(\frac{\forall}{A} - \Delta\right) \quad (+1)$$

$$\therefore \quad p = p_0 + \frac{k}{A}\left(\frac{\forall}{A} - \Delta\right) \Rightarrow p = p_0 + \frac{k}{A^2}\forall - \frac{\Delta k}{A}$$

$$\Rightarrow \quad p = \underbrace{\left(p_0 - \frac{\Delta k}{A}\right)}_{\text{constant}} + \underbrace{\frac{k}{A^2}}_{\text{constant}}\forall$$

$$\boxed{p = C_0 + C_1\forall} \quad (+1)$$

Linear relation.

To find constants C_0 and C_1, we use p and $\forall$ values at states (1) and (2):

$$100\text{ kPa} = C_0 + C_1(0.3)\,m^3$$

$$1600\text{ kPa} = C_0 + C_1(0.625)\,m^3$$

$$\Rightarrow \quad C_1(0.325)\,m^3 = 1500\text{ kPa}$$

$$\Rightarrow \quad C_1 = 4615.4\,\frac{\text{kPa}}{m^3}, \quad \therefore C_0 = 100 - 0.3 \times 4615.4$$

$$= -1284.62\text{ kPa}$$

$$\therefore \quad p = -1284.62\text{ kPa} + 4615.4\left(\frac{\text{kPa}}{m^3}\right)\cdot\forall \quad (+1)$$

If the piston is heated to $(1.4\,MPa) = \underbrace{1400\text{ kPa}}_{\text{State (3)}}$, the $\forall$ can be computed from this linear relation:

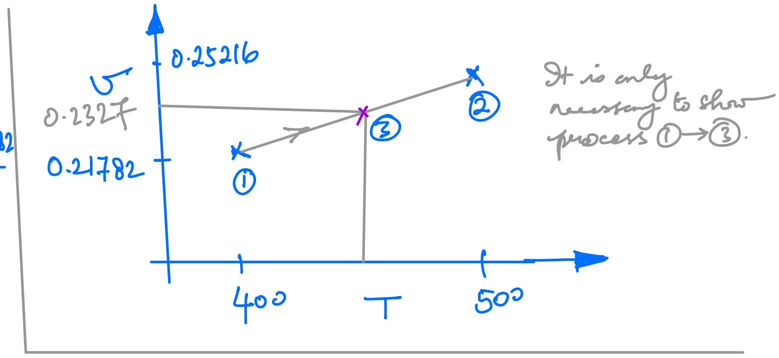
$$\forall_3 = \frac{(p_3 + 1284.62)\text{ kPa}}{4615.4\text{ kPa}/m^3} = 0.58166\,m^3$$

$$\therefore \quad v_3 = \frac{\forall_3}{m} = 0.2327\,\frac{m^3}{kg} \quad (+1)$$

at 1.4 MPa, $v_f = 0.001149\,\frac{m^3}{kg}$, $v_g = 0.14078\,\frac{m^3}{kg}$

$\because v_3 > v_g \Rightarrow$ state (3) is superheated state $(+1)$

From superheated table @ 1.4 MPa, v_3 lies b/w v values at 400°C and v at 500°C. So we perform linear interpolation.



$$\therefore \frac{0.2327-0.21782}{T-400} = \frac{0.25216-.21782}{500-400}$$

$$\Rightarrow T-400 = \frac{100\times(.2327-.21782)}{(0.25216-0.21782)}$$

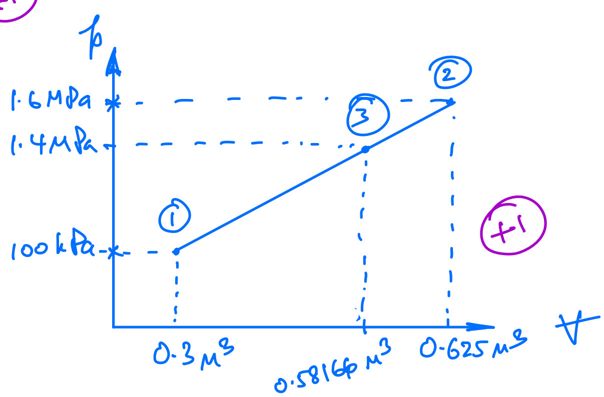
$$= 43.33$$

or, $\underline{T_3 = 443.33 \text{ K}}$ +1

Also $$\frac{u_3 - u_{400°C}}{443.33-400} = \frac{u_{500°C}-u_{400°C}}{500-400}$$

$$\Rightarrow u_3 = \left(\frac{43.33}{100}\right)\times(3121.8-2953.1) + 2953.1 \quad \text{kJ/kg}$$

$$= 3026.19 \text{ kJ/kg}$$ +1



+1

For process 1 → 3:

$$\Delta E = Q_{1\to3} - W_{1\to3}$$

$\Delta U + \cancel{\Delta KE} + \cancel{\Delta PE}$

0 0 (negligible)

$$\therefore Q_{1\to3} = \Delta U + W_{1\to3}$$

$$= m(u_3 - u_1) + \int_1^3 p\, d\mathcal{V}$$

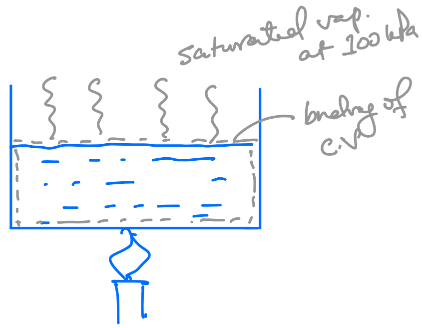
$$\Delta U = 2.5\times(3026.19 - 564.117) \text{ kJ}$$

$$= 6155.18 \text{ kJ}$$

$$W_{1\to3} = \int_1^3 p\, d\mathcal{V} = \int_1^3 (C_0 + C_1\mathcal{V})\, d\mathcal{V} = C_0(\mathcal{V}_3 - \mathcal{V}_1) + \frac{C_1}{2}(\mathcal{V}_3^2 - \mathcal{V}_1^2)$$

$$= 331.926 \text{ kJ}$$ (+1)

$$\therefore\ Q_{1\to 3} = (6155.18 + 331.926)\text{ kJ} = \underline{6487.1 \text{ kJ}}$$ (+1)

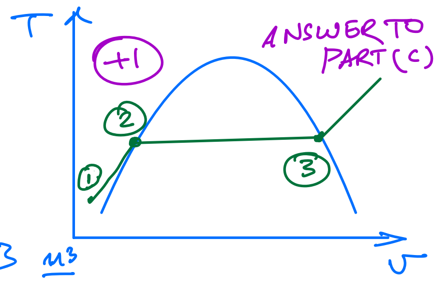


Q2

State ①: $T_1 = 25°C$ and $p_1 = 100$ kPa

Now, at $p = 100$ kPa, $\underline{T_{sat.} = 99.61°C}$

Since $T_1 = 25°C < T_{sat}(p_1)$, State ① is sub cooled liquid state (+1)



For subcooled liquid state:

$$u_1 \simeq u_f(T_1) = 104.83 \frac{\text{kJ}}{\text{kg}}, \quad v_1 \simeq v_f(T_1) = 0.001003 \frac{\text{m}^3}{\text{kg}}$$

$$\begin{aligned} h_1 &\simeq u_1 + p_1 v_f \\ &= u_f(T_1) + p_{sat}(T_1) v_f + [p_1 - p_{sat}(T_1)] v_f \\ &= h_f(T_1) + v_f(T_1) \cdot [p_1 - p_{sat}(T_1)] \\ &= \underbrace{104.83}_{\text{kJ/kg}} + \underbrace{0.001003}_{\text{m}^3/\text{kg}} \times [\underbrace{100 - 3.1698}_{\text{kPa}}] \\ &= 104.93 \text{ kJ/kg} \end{aligned}$$ (+1)

Just $h_1 \simeq h_f(T_1)$ should also be fine.

$$m_1 = \frac{V_1}{v_1} = \frac{(1\text{ L}) \times 10^{-3} \frac{\text{m}^3}{\text{L}}}{0.001003 \text{ m}^3/\text{kg}} = 0.997 \text{ kg}$$

__State ②__: End of sensible heating: $\mapsto$ saturated liquid $p_2 = 100$ kPa

$$v_2 = v_f = 0.001043 \frac{\text{m}^3}{\text{kg}}, \quad u_2 = u_f = 417.40 \frac{\text{kJ}}{\text{kg}}$$

$$h_2 = h_f = 417.51 \frac{\text{kJ}}{\text{kg}}$$ (+1)

__State ③__: State at which vapour is going out of the control volume.

$$p_3 = 100 \text{ kPa}, \quad \underline{v_3 = v_g = 1.6941 \frac{\text{m}^3}{\text{kg}}}, \quad \underline{u_3 = u_g = 2505.6 \frac{\text{kJ}}{\text{kg}}}$$

$$h_3 = h_g = 2675 \ \frac{kJ}{kg}$$ (+1)

For process (1) → (2) : Sensible heating.

(no incoming mass)

$$\Delta E = Q_{1-2} - W_{1-2} + \cancel{m_i}\left[h_i + \frac{V_i^2}{2} + g z_i\right]$$

$$- \cancel{m_e}\left[h_e + \frac{V_e^2}{2} + g z_e\right]$$

0 (no outgoing mass)

(neglect)

$$\Delta U + \cancel{\Delta KE} + \cancel{\Delta PE} = Q_{1-2} - W_{1-2}$$

or, $Q_{1-2} = \Delta U + W_{1-2}$

$$= m_2 u_2 - m_1 u_1 + p(V_2 - V_1)$$

$$= m(u_2 - u_1) + m\,p(v_2 - v_1)$$

$$= 0.997 \times [417.4 - 104.83 + 100 \times (0.001043 - 0.001003)]$$

$$= 311.64 \ kJ$$ (+1)

Rate of supply of heat = 0.9 kW

∴ Time required for sensible heating := $\frac{311.64}{0.9} \frac{kJ}{kW}$

ANSWER TO PART (a) $= 346.26\ s$ (+1)

For process (2) — (3) when entire water evaporates, During this process C.V. shrinks continuously till it vanishes. C.V. surrounds liquid portion of water. Vapor escapes this control volume when heating is done:

0 (no incoming mass) 0 0 (neglect)

$$\Delta E = Q_{2-3} - W_{2-3} + \cancel{m_i}\left(h_i + \frac{V_i^2}{2} + g z_i\right)$$

$$- m_e\left[h_e + \frac{V_e^2}{2} + g z_e\right]$$

m_1 h_3 0 0 (neglect)

$\Delta U + \cancel{\Delta KE} + \cancel{\Delta PE}$

0 0 (neglect)

$m_3 u_3 = 0$ (as $m_3 = 0$)

pressure is const during process.

$$\Rightarrow \cancel{U_3} - U_2 = Q_{2-3} - p(V_3 - V_2) - m_1 h_3$$

$$\Rightarrow Q_{2-3} = p(V_3 - V_2) + m_1 h_3 - m_2 u_2$$

$$= 100(0 - 1 \times 10^{-3}) + 0.997 \times 2675 - 0.997 \times 417.4 \ kJ$$

$= 2250.73 \text{ kJ}$ +2

Time required for water to completely vaporize $= \dfrac{2250.72 \text{ kJ}}{0.9 \text{ kW}} = \underline{2500.88}$ +1

ANSWER TO PART (b)